

```
using System;

using System.Collections;

using System.Collections.Generic;


// Struct representing Book details
public struct Book
{
    public string Title { get; set; }
    public string Author { get; set; }
    public int CopiesAvailable { get; set; }


    public Book(string title, string author, int copies)
    {
        Title = title;
        Author = author;
        CopiesAvailable = copies;
    }
}


// Base class
public class Library
{
    public string LibraryName { get; set; }
    public List<Book> Books { get; set; } = new List<Book>();


    public Library(string name)
    {
        LibraryName = name;
    }


    // Virtual method to allow overriding
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public virtual void DisplayLibraryInfo()
{
    Console.WriteLine($"Library: {LibraryName}");
    Console.WriteLine("Books available:");
    foreach (var book in Books)
    {
        Console.WriteLine($"Title: {book.Title}, Author: {book.Author}, Copies:
{book.CopiesAvailable}");
    }
}

// Operator Overloading to merge two libraries
public static Library operator +(Library lib1, Library lib2)
{
    Library mergedLibrary = new Library(lib1.LibraryName + " & " + lib2.LibraryName);
    mergedLibrary.Books.AddRange(lib1.Books);
    mergedLibrary.Books.AddRange(lib2.Books);
    return mergedLibrary;
}

// Derived class demonstrating inheritance
public class Member : Library
{
    public string MemberName { get; set; }
    public int MemberID { get; set; }

    public Member(string libName, string memberName, int memberID) : base(libName)
    {
        MemberName = memberName;
        MemberID = memberID;
    }
}

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    }

    // Overriding base class method
    public override void DisplayLibraryInfo()
    {
        base.DisplayLibraryInfo();
        Console.WriteLine($"Member: {MemberName} (ID: {MemberID})");
    }
}

// Sealed class (cannot be inherited)
public sealed class Librarian
{
    public string LibrarianName { get; set; }

    public Librarian(string name)
    {
        LibrarianName = name;
    }

    // Sealed method (cannot be overridden)
    public sealed void ManageLibrary()
    {
        Console.WriteLine($"Librarian {LibrarianName} is managing the library.");
    }
}

// Extension Method for List<Book>
public static class LibraryExtensions
{
    public static void ShowBookTitles(this List<Book> books)

```

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{
    Console.WriteLine("Book Titles:");
    foreach (var book in books)
    {
        Console.WriteLine(book.Title);
    }
}
}

```

class Program

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{
    static void Main()
    {
        try
        {
            // Boxing: Storing an int inside an object
            int copies = 5;
            object boxedCopies = copies;

            // Unboxing: Extracting the value back to int
            int unboxedCopies = (int)boxedCopies;

            Library lib1 = new Library("Central Library");
            Library lib2 = new Library("City Library");

            lib1.Books.Add(new Book("C# Programming", "John Doe", 10));
            lib2.Books.Add(new Book("Data Structures", "Jane Smith", 5));

            // Demonstrating operator overloading
            Library mergedLib = lib1 + lib2;
            mergedLib.DisplayLibraryInfo();
        }
    }
}

```

```
// Creating Member instance

Member member = new Member("Central Library", "Alice", 101);

member.Books.Add(new Book("AI & ML", "Andrew Ng", 3));

member.DisplayLibraryInfo();


// Using Extension Method

member.Books.ShowBookTitles();


// Using System.Collections

ArrayList mixedCollection = new ArrayList();

mixedCollection.Add(new Book("Cyber Security", "Bruce Schneier", 4));

mixedCollection.Add(100); // Demonstrating storing multiple data types


// Sealed class instance

Librarian librarian = new Librarian("Mr. Watson");

librarian.ManageLibrary();

}

catch (Exception ex) // Exception Handling

{

    Console.WriteLine("Error: " + ex.Message);

}

}

}
```